FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 03

AI Project - Assignment 3

THEME:

SMART CITY - ECO-ZONE AIR CONDITIONER

LECTURER: DR. NORSHAM BINTI IDRIS

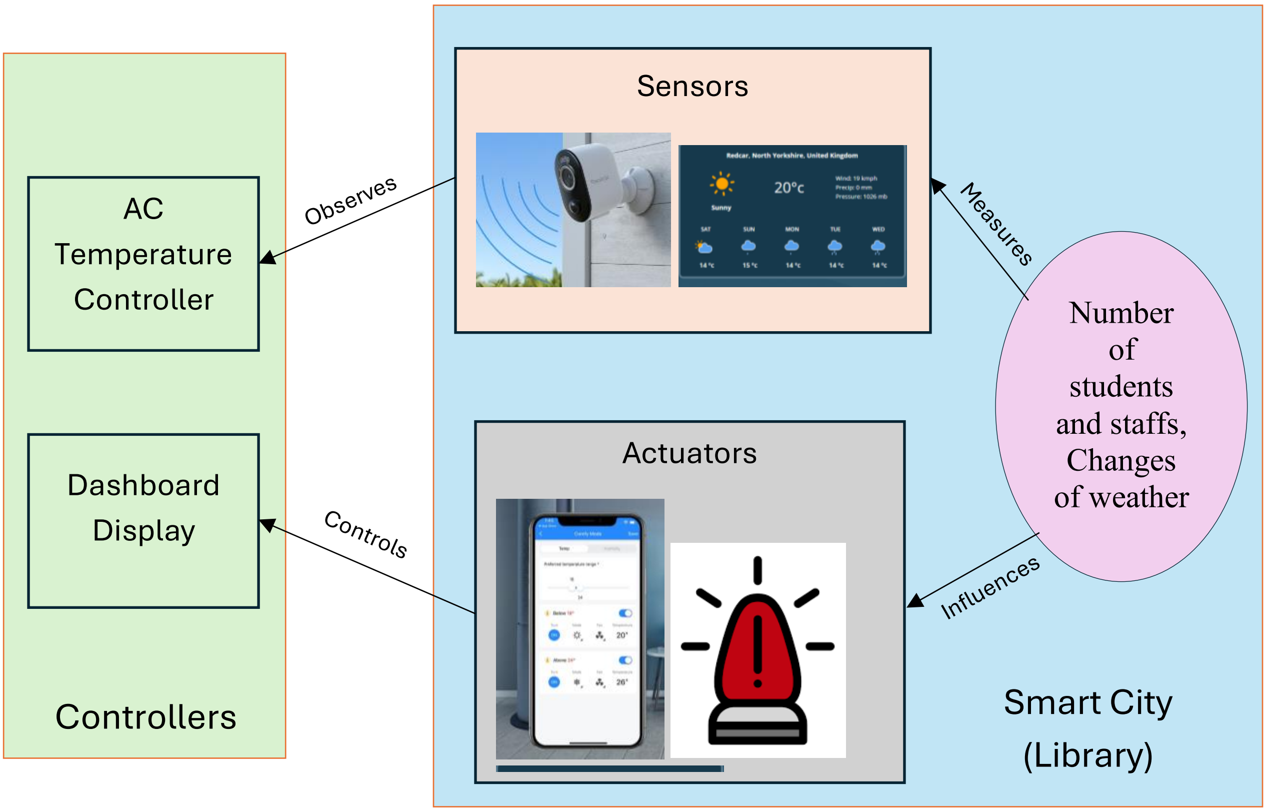
STUDENT NAME	MATRIC NO
CHAU YING JIA	A23CS0213
NURUL ASYIKIN BINTI KHAIRUL ANUAR	A23CS0162
NEO LI XIN	A23CS0253
AFIF SHAQIR IRFAN BIN ARQAM	A23CS0204

Table of Contents

1.0 PEAS Model

Performance	User comfort, Energy efficiency, Cost saving, Response time
Environment	Library Zones (Discussion room, Open area, Private study room), External Weather (Sunny, Rain), Students and Library Staffs
Actuators	Dashboard display, Status alert system
Sensors	Motion sensors, cameras, Internal Thermostat, External Weather API

2.0 Relations Between Each Property



3.0 Explanation in Proof of Concept

Performance

The AI-based air conditioning system is intended to improve user comfort while optimizing energy consumption in the library environment. The system makes intelligent cooling decisions based on occupancy levels, time context and temperature conditions. As a result, when library areas are crowded, the system detects the need for lower temperatures to maintain a comfortable study atmosphere by lowering the temperature. Furthermore, during low-occupancy times or at night, the system identifies opportunities for energy saving modes by recommending higher temperature settings. The system assures efficient cooling operations, reduces energy waste and illustrates possible cost savings while maintaining a conducive learning environment by implementing rule-based decision making.

Environment

The Proof of Concept setting includes several library locations such as open study spaces, discussion rooms and individual study rooms. Each of these has a specific usage pattern. The system detects differences in these areas by using simulated data from occupancy and temperature sensors as well as time. By handling each zone separately, the AI may detect zone's specific temperature requirements rather than using a single setting for the entire library. The environment also takes consideration for the external weather since variations in the outside temperature can impact the requirement for indoor cooling. This environment enables the system to respond dynamically to user presence and environmental changes, thereby promoting energy efficiency, user comfort and sustainable library management.

Actuators

The actuators used in the AI-based air conditioning system are dashboard display and status alert system. The primary function of the dashboard is to allow library staff to check real-time air temperature in each zone as well as manually control the air conditioners in the library. This reduces the workload of library staff as they can easily monitor and adjust each floor's air conditioner through the dashboard. Next, the status alert system's function is to actively notify the staff when there are adjustments required. For instance, if the number of users in a zone exceeds the limits, the system will alert the library staff through a pop-up notification asking if they want to adjust to cooler temperature. The staff then uses the dashboard to adjust the air conditioner temperature accordingly based on the state space graph.

Sensors

The input data from the sensors will be passed into the AI logic module and be carried out by the actuators. The types of sensors are motion sensors, cameras, internal thermostat and external weather API. Motion sensors help to detect library users entering or leaving; cameras help to monitor the number of students in each zone; internal thermostat checks temperature for autonomous air conditioner adjustments; and external weather API to read temperature outside the library. These sensors are essential to ensure the flow of FOL so that the AI system can automatically control the air conditioners in the library.

4.0 Behaviours of Agent to Achieve the System Goal

The intelligent agent in the Smart City Eco-Zone Air Conditioner system works by constantly observing what is happening in the library and responding to it. The main goal is to maintain comfortable environment for users while reducing unnecessary energy consumption. To achieve this, the agent continuously monitors each library zone, makes decisions based on real-time conditions, and then triggers appropriate actions.

Firstly, the agent will go through monitoring phase which the agents will collect information from various sensors. One of the sensors are motion sensors and cameras which will detect how many users are in each area. Other than that, the Internal thermostat readings show the current temperature inside each zone and external weather API provides outdoor temperature and weather conditions.

After that, the agent will continue decision-making phase after receiving the data, the agent evaluates whether the environment is comfortable or if changes are needed. It uses a set of rules to guide decision. The rules used are if there are many users in a zone (for example, more than 10 people), the system lowers the temperature to keep the space comfortable. Other than that, if a zone has very few users or is empty, the system increases the temperature or switches to energy-saving mode. In addition, when the outside weather is hot, the system anticipates higher cooling needs and if the indoor temperature goes beyond the comfort level, cooling is increased. Moreover, during late night or low usage periods, the agent prioritizes saving energy instead of strong cooling. This decision helps the agent to adjust dynamically instead of using one fixed temperature for all situations.

The agent then will continue to action phase once a decision is made, the agent activates the actuators or carry out the action. Firstly, the dashboard display shows real-time updates for each library zone and then alert the system to notifies library staff if a change is needed. Staff then can then adjust the air conditioner settings through the dashboard based in agent’s recommendation.

To achieve the system goals, the agent will go through this continuous cycle of monitoring, deciding, and acting. This will keep the library comfortable when many people are present and reduce electricity usage when areas are empty or not actively used. In addition, the agent able to respond quickly to crowd changes and temperature changes and support a smart and sustainable library environment by balancing comfort and energy efficiency.